

Making Sense of Online Discussions: Can Automated Reports help?

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ABSTRACT

Enabling healthier online deliberation around issues of public concerns is an increasingly vital challenge in nowadays society. Two fundamental components of a healthier deliberation are: i. the capability of people to make sense of what they read, so that their contribution can be relevant; and ii. the improvement of the overall quality of the debate, so that noise can be reduced and useful signals can inform collective decision making. Platform designers often resort to computational aids to improve these two processes. In this paper, we examine automated reporting as promising mean of improving sensemaking in discussion platforms. We compared three approaches to automated reporting: an abstractive summariser, a template report and an argumentation highlighting system. We then evaluated improvements in sensemaking of participants and the perception on overall quality of the debate. The study suggests that argument mining technologies are particularly promising computational aids to improve sense making and perceived quality of online discussion, thanks to their capability to combine computational models for automated reasoning with users' cognitive needs and expectation of automated reporting.

CCS CONCEPTS

• **Networks** → Online social networks; • **Computing methodologies** → *Natural language generation*; • **Information systems** → *Collaborative and social computing systems and tools*.

KEYWORDS

automated reporting, online discussion systems, sensemaking, NLP, argumentation mining, automatic text summarisation

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1 INTRODUCTION

Deliberation is the careful discussion before decision, and it can be defined as the thorough dialogical assessment of the reasons for and against a measure before a decision is made. Distributed online deliberation is increasingly needed, as further confirmed by the ongoing COVID-19 crisis. We are currently witnessing the continuous transfer of face-to-face societal functions in all manner of organisations, such as industry, education and government, to digital setting. Many companies are talking of making some of these changes permanent, which means that in the coming future employees may be increasingly asked to absolve to their activities from home, and with the support of digital technologies. Social media and online discussion platforms are commonly used to facilitate group discussion in such contexts. Nonetheless, these technologies have well recognised limitations and threats to healthy group deliberations, such as degraded quality of discussion [17], increased polarisation [35], echo chambers [20] -or information cocoons- [15], up to provoking mental depression [17]. Furthermore, as people tend to follow opinions of like-minded others, weakly designed deliberation/discussion platforms can be exploited for distribution of biased and unsupported information, rumors, misinformation, speculations and other dishonest behaviours [10]. This seriously undermines the potential of these technologies to support effective discussion-based collaboration.

This motivates our research into new technological support for online deliberation, our project specifically looks at overcoming two core limitations of existing systems: lack of overall discussion quality, specifically in term of data structure and evidence-based reasoning; and lack of features to support sensemaking and situational awareness for people to meaningfully participate in the online discussion. This article presents preliminary work looking at computational aids to improve a key stage of participants sense-making that is: communicating results and story-tell [31]. We start from the premise that automated reporting can tell a "story" of the state and progress of an online debate, which can improve participants understanding and positively affect the overall quality of the debate by helping them to better inform their contribution. As people struggle to comprehend large or dynamic debates, computational aids such as a automatic summariser or a text-highlighter of argumentative structures, have the potential to improve the current status of online debate technologies by enabling participants to quickly, but systematically, grasp the overall state of a debate before or while they are contributing. This may lead to better quality debates, for instance by enabling people that are new to the discussion to inform where they could better contribute. Automated reporting can be also particularly useful in highly dynamic debates, which tend to evolve fast, for returning participants to quickly grasp

the progress made since they last contributed, thus reducing idea repetition.

This study looks specifically at the capability of automated reports to improve newcomers' sensemaking and perceived quality of debate. We compared three different automated reporting mechanisms: Abstractive summarisation, Templated summary and Argumentation highlight - excerpts of each is shown in figure 1. We aimed at understanding which one of the three approaches is more promising in improving new participants' capability to make sense of an online discussion. We also aimed to investigate the extent to which providing a summary can improve newcomers' perception of the overall debate's quality. To this end, we carried out a cross-sectional between subjects quantitative study where we compared 4 different groups: (3 treatment groups), each with a different automated reporting method and one control group (deliberation with no automated report).

Results were evaluated against a theory-based set of Sensemaking (SM) and Quality of Debate (QoD) features (Figure 2). The above quantitative study was followed by a focus group to discuss with users each automated reporting method and to qualitatively assess their applicability in the context of a deliberation platform. In the following, we motivate and describe the research, and draw preliminary conclusions which can inform the design of future automated reporting systems for online discussion.

Our findings are currently being used to directly inform the design of a novel online deliberation platform, BCAUSE¹, which aims to improve current social media and online discussion platform by providing three main innovations: (i) Low cost argumentation structuring, (ii) Advanced Visual analytics and (iii) decentralised anonymous reputation. This study provides the building blocks for the design of the automated reporting engine of the BCAUSE platform.

2 RELATED WORK

A challenge frequently faced by participants of online deliberation platforms is to deduce the reasoning (the "why?") of the arguments presented. There is a fundamental difficulty of *making sense*, or in other words to build an accurate internal structured representation of the facts, evidence and opinions presented. Deliberation platform designers often attempt to assist this *sensemaking* process by providing tools for users to manually map the argumentative structure of a debate [8], however, despite their recognised advantages, manual argument mapping solutions are hard and lengthy to produce and come with an extra cost, as they require a cognitive leap for users to engage efficiently in the deliberation process [33]. As an alternative to these approaches automated reporting aims to provide a light touch support to sense making of the debate, in a linear narrative form, therefore much easier for users to digest and react to.

Automated reporting has been extensively investigated in the HCI and CSCW community with applications in different areas varying from learning [12, 14, 29], to collective intelligence [19, 27, 36, 38], to computational journalism [1, 25] and citizen science [22, 37]. Its overall goal is to improve comprehensibility of a vast amount of data or to ease complex cognitive processes with means

of computational tools. This comes in different flavours as to improve participation in communities, enable users to make better use of chat [41], to provide manageable workflows in complex crowdsourcing environments, e.g. [6], or provide assistance to moderation systems, e.g. [5, 23, 28].

Its main function is to provide a quick overview of the state and progress of a discussion. Different platforms employ different approaches. For example, Wikum [42] proposes a collaborative interaction workflow of summarisation, relying on crowdsourcing for SM like CrowdIA [24]. This approach requires manual summarisation, and may therefore cope less well with large scale discussion contexts with many participants and lengthy contributions. Visualisation is also a popular mechanism of displaying summarised information, e.g. Kialo [3] displays a taxonomy of the arguments and counter-arguments as sunburst diagram, while Consider.It [34] displays the consensus of participants in an agreement axis. Whereas more suitable to convey results of large debates, visual analytic approaches are harder to make sense of for non-expert users, and often provide overwhelming information.

In this paper we focus on Natural Language Processing (NLP) approaches to generate a quick and intuitive textual report of an online discussion, which can provide a scalable and accessible solution to summarise the state and progress of an online debate. In particular we focus on three main approaches to automatic reporting of the online discussion, each showing particular promise in their own field: Natural Language Generation (NLG) -the automatic generation of natural language text from structured data- which has a growing interest from industry [30]; *abstractive* summarisation - in which the important subset of sentences included in the summary are detected and paraphrased to generate the output summary (e.g. [32, 40]); and argument mining - which extract the argumentation structure in a given text by identifying participants rhetorical moves within the debate, in form of claims and arguments [18].

In the following we present a study in which we compared these three automated reporting approaches, in the context of a simple argumentation-based discussion platform (such as Kialo.com and DebateHub.net), with the objective to find out which automated reporting approach lays most promise to enhance new participants' SM and improve their perception of debate's quality.

3 RESEARCH QUESTION AND EXPERIMENTATION DESIGN

Our overall assumption and research hypothesis is that a textual summary presented to a new user in advance or in parallel to his interaction with a discussion topic improves his understanding of the overall state of the debate and enables healthier participation in it. This assumption drives the research questions below:

RQ1: To what extent automated reports can improve newcomers' personal SM and overall perception of the QoD?

RQ2: What automated reporting approaches show the biggest improvements, and what SM and QoD factors they affect?

In order to address these research questions, we compared three automated reporting methods in the attempt to establish which one would be more suitable to achieve our goals. For the comparison we used a mixed methods approach with explanatory sequential

¹<https://bcause.kmi.open.ac.uk/>

(i) Abstractive summary

- Students do not need government's funding (aka loans and grants) because they can learn for free (with open courseware) and get college credits for low costs through massively open online courses (mooc's) .
- alternatives may not be as easily accessible as public universities
- in most countries there is no right to study other than the right not to be hindered by the state
- automation and robotics will reduce the need for uneducated work force . more and more workers need a basic education and skills they get from further education
- children from a higher class background doesn't have more opportunity through the education system . inequalities would have been entrenched in primary and secondary schooling by then • general public education improves the employ-ability of all youth beyond basic knowledge

(ii) Templated summary

This debate has drawn 646 contributions from 110 participants. There have been many claims centering mainly around 15 ideas. Some of the strongest ideas in favor of this proposition are that everyone has the right to study and educated, while those opposing the idea claim that the cost of attending makes students more accountable to their work. Also, the need for a college degree has increased, however, students do not need government's funding (aka loans and grants) because they can learn for free (with open courseware) and get college credits for low costs through massively open online courses (mooc's) . alternatives may not be as easily accessible as public universities

(iii) Excerpt of Argument (claims or evidence) highlight report**claim - evidence**

Students do not need the government 's funding (aka loans and grants) because they can learn for free (with open courseware) and get college credits for low costs through massively open online courses (MOOC 's). General public education improves the employ-ability of all youth beyond basic knowledge. In 1988, nominal median household income was \$ 27,225 with average college tuition for a public four-year university \$ 3190 per year ... More and more workers need a basic education and skills they get from further education . Considering the tuition compared to income, it has only had a 38.4 percent real increase in 30 years when compared to median household income, resulting in a 1 percent real increase year-over-year when compared to median household

Figure 1: Examples of automatic generated reports on the debate topic : “Higher education should be publicly funded.” using (i) abstractive summary, (ii) templated summary and (iii) argument highlight

design (two phase design)[7] consisting of a quantitative study to identify results for a follow-up investigation, in phase two, which is then carried out with a qualitative data analysis.

3.1 Design

We focused on fully automated reporting methods and chose for evaluation three methods that are operable on textual data and have the potential to unveil rhetorical structures within text, which makes them more suitable for the analysis of discussion data:

- *Abstractive summarisation*: In contrast to the simpler *extractive summarisation*, it does not just extracts significant snippets from the original, but rather generates an entirely new text [32]. It relies on advanced NLP techniques to identify lexical and discourse elements of the input text producing an intermediate semantic representation of it, before proceeding to synthesise important bits of information to newly created document. Recent advancements with the use of encoding transformers have produced models that achieve close to human performances ([13, 40, 43]). The abstractive summary that we used was generated with the model described in [32] and an excerpt can be seen in figure 1(i).
- *Templated NLG summary*: As the simplest form of Natural Language Generation, templated summaries are systems that are based on static or dynamic templates, a blend of human-crafted rules to “fill-in-the-blank” in a text template. An advantage of templated NLGs would be that they can easily combine heterogeneous data sources (and not just rely on the pure textual input), for example in our case we included some simple statistical indicators (number of participants in the overall debate, number of claims, number of total contributions). The template used in our study was realised using SimpleNLG² engine and can be seen in figure 1(ii).
- *Argumentation highlight*: is a different kind of reporting, aimed at highlighting and revealing argumentation structures in the given text. It relies on Argumentation mining

[18], a NLP method capable of analyzing people's argumentation, i.e. extracting arguments and their relations from text [4]. An argument mining processed version of the debate (where arguments, claims and evidence are highlighted) can be seen in figure 1(iii). The claim and evidence extraction used in this study was done using the MARGOT system [26]).

4 QUANTITATIVE COMPARATIVE STUDY

The objective of the quantitative study was to systematically compare the three automated reports described above in terms SM and perceived QoD improvements. As mentioned in section 2, quantifying *Sensemaking* to measure it as an experimental variable is a challenging task. For our study we used Alsufiani's [2] theoretical factors of SM, with an extra feature to assess improvements in Reflection, as proposed by [9]. This resulted in the 9 SM factors presented in figure 2. Similarly, for measuring *Quality of Debate* we used a merged set of 11 factors proposed in [16] and [21] to construct a questionnaire to quantitatively assess users' perceived QoD (also presented in figure 2).

4.1 Hypotheses

In order to address the **RQs** we formed the following testable hypotheses: H_1 : Any form of report improves internal sensemaking of participant and perception of the debate quality compared to no summary present. H_2 : A templated summary report improves SM and perception of QoD compared to an abstractive summary report. H_3 : An argument highlight report improves SM and perception of QoD compared to a templated summary or abstractive summary reports. Our experimental conditions consists of the presence (or not) of an automated report, with the following four conditions: A =control/no report, B =abstractive summary report, C =templated report and D =argumentation highlight.

4.2 Task

Users were presented with a a simple user interface where an issue was debated with pro and con arguments and an automated report

²<https://github.com/simplenlg/simplenlg>

Code	Literature sensemaking factor	Adapted definition
SM1	[41] Retrospect	Reflection
SM2	[2] Gaining insight	Insights
SM3	[2] Finding connections	Focus
SM4	[2] Structuring	Argumentation
SM5	[2] Reducing confusion,	Explanation
SM6	uncertainty, and ambiguity	Assess Facts and Evidence
SM7		Distinguishing
SM8	[2] Gap-finding and	Assess assumptions
SM9	gap-bridging	Change assumptions

Code	Literature quality of debate factor	Adapted definition
QoD1	[21] Message clarity	Message clarity
QoD2	[21] Message quality	Message quality
QoD3	[21] Presence of newknowledge	New knowledge
QoD4	[21] External resources used	External resources
QoD5	[21] Resolution of discussion	Discussion resolution
QoD6	[35] Reasoned opinion expression	Reasoned opinions
QoD7	[35] Sourcing	Trustworthy evidence
QoD8	[35] Distinct views - disagreement	Distinct opinions
QoD9	[35] Equality	Equality
QoD10	[35] On topic	On topic
QoD11	[35] Engagement	Engagement

Figure 2: Sensemaking and Quality of Debate evaluation factors

was presented along according to the given condition forming four different groups correspondingly. The issue discussed was “*Should higher education be publicly funded?*” and data were scraped from the kialo.com³ platform. Only the top 10 pro and top 10 con arguments were presented. The ranking was assigned according to the given order by Kialo platform which is deduced as a mixture of voting and user rating.

Users were then asked to devote at least five minutes to study, read and comprehend the issue discussed and arguments of both sides. As a completion criterion users were asked to (i) Identify the most significant (strongest) argument in their opinion, (ii) Contribute at least one argument for, or against, and (iii) Summarise their understanding of the debate in a short text of 50-100 words. We assigned for each group 40 participants (160 total) and used Mechanical Turk to recruit users. A reward of equivalent 12\$ per hour was given for each completed task and users had 10 min to complete the task. For each task to be considered accepted, (i) a summary of more than 50 words, (ii) authentic style (not just copy paste a long piece of text from the platform) and (iii) justification of answers consisting of more than five words was required.

4.3 Analysis

We used discrete variables to measure participants’ SM and perceived QoD (likert scale survey answers). We mapped the likert scale responses to a linear scale paying attention to reverse the negated questions. An initial Shapiro-Wilk Test showed that the values do not conform to a normal distribution and the variance is not homogeneous, therefore, we used a non parametric test to compare the conditions. We chose a Kruskal–Wallis test [39], also known as one-way ANOVA on ranks, followed by a post hoc examination using Dunn’s test [11] to reveal which group was stochastically dominating.

We present the results of Kruskal-Wallis H test (df=3, $\alpha=0.05$) of quality and SM features in table 1 and 2 containing a complete overview of means and standard deviations by condition, the Kruskal-Wallis test statistic (H-stat) and corresponding p-value of each feature. Our analysis of SM features showed a significant difference ($p<0.05=\alpha$) between the four conditions for Reflection (SM1), Insights (SM2), Focus (SM3) and Argumentation (SM4) while no significant differences * emerged for the rest of SM features (SM5-SM9). For the QoD features, a significant difference was observed for

Message Clarity (QoD1) and Trustworthy evidence (QoD7) feature. The subsequent pairwise comparisons in the statistically significant features are shown in table 3, where we used Bonferroni-adjusted significance level of $\alpha^* = 0.05/6 = 0.00833$.

We can confirm hypothesis H_1 for the argumentation highlight report (group D), which outperforms the control group for better Message Clarity (QoD1), Reflection (SM1), and Argumentation (SM4) features. This confirms our assumption that argumentative technologies perform better regarding key SM features. We cannot draw a conclusion however on the effects on the remaining SM factors such as reducing confusion, uncertainty and ambiguity.

Hypothesis H_2 cannot be confirmed i.e. we cannot draw a conclusion on which type of summary based report (abstractive or template) performs better in any of the features examined. Finally, we can confirm H_3 , that Argumentation Highlight performs better than abstractive summary, by providing more Insights (SM2), and performs better than template report in Message Clarity (QoD1), Insights (SM2), Focus (SM3) and Argumentation (SM4).

Argument mining techniques are explicitly designed with the goal to improve discourse analysis, but their impact in realistic application contexts is yet to be demonstrated. Our preliminary findings confirm that argument mining results, when presented along an online discussion, even in the simplest form (as basic highlights) perform better than other automated reporting approaches, and can help users to better make sense of the logic structure of a debate, gain insights and perceive a clearer message.

5 QUALITATIVE STUDY

A small focus group was carried out to explore the value and applicability of automated reporting in online deliberation systems and investigate further the qualitative differences between the methods compared in the quantitative study. Focal point of the investigation was around the SM and QoD features that did not reveal any statistical significance in the quantitative study. We conducted the focus group at the university’s premises in a meeting room equipped with external monitors to display the different reporting interfaces alongside the online discussion. The focus group lasted one hour and gathered views from three participants with diverse backgrounds (Computer Science, Education and Biology). We asked exploratory questions regarding the qualitative characteristics of each of the automated reports (B, C, D) and asked participants to discuss, compare, identify shortcomings and suggest improvements.

³The full topic can be found at <https://www.kialo.com/should-higher-education-be-publicly-funded-7565?path=7565.0~7565.1>

	SM1 Reflection	SM2 Insights	SM3 Focus	SM4 Argumentation	SM5 Explanation	SM6 Assess facts	SM7 Distinguishing	SM8 Assess assum.	SM9 Change assum.
A (control)	3.925 (0.858)	3.675 (1.022)	3.25 (1.334)	3.5 (1.062)	3.25 (1.295)	2.75 (1.295)	3.85 (0.863)	3.3 (1.159)	2.95 (1.131)
B (abs. sum.)	4.325 (0.944)	3.45 (1.108)	3.6 (1.354)	3.925 (1.022)	3.475 (1.377)	2.55 (1.376)	4.05 (1.011)	3.55 (1.338)	3.2 (1.181)
C (temp. rep.)	4.05 (0.714)	3.5 (0.784)	3.1 (1.194)	3.5 (1.012)	3.125 (1.158)	2.875 (1.158)	3.75 (0.954)	3.35 (1.098)	3.125 (1.244)
D (arg. high.)	4.35 (0.921)	4.05 (1.011)	3.875 (1.244)	4.15 (1.026)	3.575 (1.393)	2.4 (1.373)	4.175 (0.812)	3.725 (1.395)	3.25 (1.255)
H-stat(3)	12.036	11.271	9.960	12.800	3.492	3.635	6.014	4.659	1.785
p-value	0.0072	0.0103	0.0189	0.005	0.3217	0.3036	0.1108	0.1985	0.618

Table 1: Kruskal-Wallis statistical test of Sensemaking features

	QoD1 Msg Clarity	QoD2 Msg quality	QoD3 New knowledge	QoD4 Ext. resources	QoD5 Resolution	QoD6 Reasoned opinions	QoD7 Trustworthy	QoD8 Distinct opinions	QoD9 Equality	QoD10 On Topic	QoD11 Engagement
A (control)	3.525 (0.997)	2.625 (1.102)	3.5 (1.012)	3.2 (1.114)	2.45 (1.06)	3.35 (0.892)	3.325 (0.729)	3.225 (0.973)	3.25 (1.006)	3.725 (0.784)	3.675 (0.997)
B (abs. sum.)	4 (0.905)	2.425 (1.318)	3.575 (1.129)	3.35 (1.026)	2.575 (1.152)	3.65 (0.975)	3.8 (0.966)	3.125 (1.158)	3.325 (1.227)	3.975 (0.861)	3.8 (0.992)
C (temp. rep.)	3.675 (0.858)	2.75 (1.235)	3.525 (0.846)	3.275 (0.986)	2.65 (1.051)	3.7 (0.822)	3.45 (0.875)	3.125 (0.911)	3.2 (1.042)	3.8 (0.992)	3.725 (0.876)
D (arg. high.)	4.15 (0.975)	2.125 (1.09)	3.7 (1.181)	3.125 (1.284)	2.625 (1.371)	3.85 (0.921)	3.775 (0.973)	2.975 (1.229)	3.125 (1.399)	3.85 (1.051)	3.65 (1.188)
H-stat(3)	12.426	6.818	1.535	0.508	0.679	6.187	8.384	1.002	0.493	2.34	0.442
p-value	0.006	0.0779	0.6741	0.917	0.8779	0.1028	0.0386	0.8007	0.9202	0.5047	0.9314

Table 2: Kruskal-Wallis statistical test of Quality of Debate features

Hypothesis	Group comparison	QoD1 Message clarity	QoD7 Trustworthy evidence	SM1 Reflection	SM2 Insights	SM3 Focus	SM4 Argumentation
H_1	A < B	z=2.039, p=0.041	z=2.542, p=0.011	z=2.593, p=0.009	z=0.879, p=0.379	z=1.287, p=0.198	z=1.819, p=0.068
H_1	A < C	z=0.32, p=0.748	z=0.878, p=0.379	z=0.404, p=0.685	z=1.246, p=0.212	z=0.629, p=0.529	z=0.076, p=0.938
H_1	A < D	z=3.039, p=0.002	z=2.209, p=0.027	z=2.682, p=0.007	z=1.825, p=0.067	z=2.243, p=0.024	z=2.917, p=0.003
H_2	B < C	z=1.719, p=0.085	z=1.664, p=0.096	z=2.188, p=0.028	z=0.367, p=0.713	z=1.916, p=0.055	z=1.896, p=0.057
H_3	B < D	z=0.999, p=0.317	z=0.333, p=0.738	z=0.088, p=0.929	z=2.704, p=0.006	z=0.955, p=0.339	z=1.098, p=0.272
H_3	C < D	z=2.719, p=0.006	z=1.331, p=0.183	z=2.277, p=0.022	z=3.071, p=0.002	z=2.872, p=0.004	z=2.994, p=0.002

Table 3: Dunn’s test post-hoc examination - pairwise comparisons

We transcribed the full length of the focus group and proceeded to analyze the text using the controlled vocabulary of the SM and QoD features described in related work section (deductive coding). On a second round of coding we identified six main themes, from which we can deduce the following preliminary criteria for the design of automated reports of online discussion: (i) *Accuracy*: as false claims confuse users, low accuracy shall be mitigated (i.e. shown with caution/disclaimer) e.g. P2: “[the report] says that the blue is evidence,...but sometimes is quite clearly just an opinion”, (ii) *Credibility*: presenting credible and trustworthy information to the end user is the main driver to build trust, unsubstantiated information is better to be completely ignored, e.g. P1: “[...] even if Pearce Morgan tells you the sun comes from the East and sets in the West you would be ‘Emmm’”, (iii) *Intuitive navigation*: the expectation that each piece of information is explorable is important for the user to build his internal structures, on the other hand, broken navigation is damaging users’ capability to gain insights, e.g. P2 complains: “[report D] is a bit chaotic [...] but is interesting to see things highlighted”, (iv) *Evidences*: the presence of hard evidence (e.g. scientific) improves the strength of the arguments presented, e.g. P2 emphasises: “I like the idea of highlighting the evidence [...], actually presenting in here actual undeniable evidence is part of the debate, I think it would be very very strong.”, (v) *Informative*: a good report should go beyond just repeating the input text in a shorter form but provide analytical insights on the status, progress and limitations of the ongoing debate - P3 when comparing summary approaches indicates that “It is a bit terrifying when I see such a big text, it is overwhelming. So having like four lines and understanding those instead- or having seven bullet points is more helpful”, (vi) *Structured*: each piece of the report should clearly denote the semantic relatedness to the overall debate, unveiling the

overall structure of the ideas and opinions presented, helping the SM process of structuring (SM4). P1 summarises this: “you need to show the limitations of the debate, the polarisation, political bias.. if is equal, if is diverse, if is not dominated by one opinion...”.

6 DISCUSSION

The findings of both studies corroborate our main research premise (RQ1), that automated reports have a positive effect on the internal SM of newcomers and perception of the QoD. The main trigger to the SM process, we believe is the affordance of automated reports of providing a “quick glance” to the user; offering a manageable amount of information while guiding him to the points of interest. This can be vital to newcomers (or late returning users) to find connections, gain insights, reflect on the information given, and ultimately better *make sense* of the debate.

Regarding our second research question -which of the examined automated reports improves SM and QoD the most- we observed that argumentation highlight outperforms the other two -summary based- types of report. We believe that this is due to its affordance of distinguishing between claims, arguments and evidence, helping people shape their opinions on fairer grounds, a crucial need for users. Surely, the interaction -the navigational elements- of the system tested was diminished, as we tested the debate on a basic discussion interface, to isolate other experimental factors that could skew our results. Evidently, the full integration of such reports in a fully developed online discussion platform consists of a harder design challenge which will be the focus of future research and development.

For some of the quality elements of a good debate, e.g. resolution of discussion or engagement, and the SM elements of reducing confusion, uncertainty and ambiguity we did not observe significant

effects of the presence of automated reports. Similarly, message quality and presence of new knowledge in the text, as well as equality of participation, and engagement in the discussion are wider design challenges that cannot be addressed by a simple summary, they require future research on new user interaction paradigms, and design solutions that specifically target them.

We also recognise that our study has limitations. Firstly, we took the assumption that SM and QoD variables are depended and examined them together. Also, the quantitative study has some shortcomings as that only one debate was tested - various debate topics of different thematic categories would have abated any bias coming from the choice of topic. Furthermore, the study design was based on summaries of equal length without investigating the relevance of length and other variables (e.g. amount of arguments, the structure of those) and how those influence other variables (e.g. participation and decision making) which can be examined in future studies. Finally, the virtual execution of the study did not allow for follow up questions. This was the main driver to setup a focus group afterwards, though we need to acknowledge that it had a small number of participants. Nonetheless, via our quantitative study, we confirmed our hypothesis (H1, A<D) that argument mining approaches to automated reporting, significantly improve participants SM, by increasing reflection (SM1) and idea structuring (SM4), and enhance perception of QoD, by improving message clarity (QoD1). We have also shown realtive improvements of this method compared to other automated summary approaches. Our qualitative examination also revealed that accuracy is a weakness of such systems, that originates from the low performance of the underlying NLP system. Though NLP has seen tremendous growth in the recent years, is yet not perfect; actually the tasks of automatic summarisation and argumentation mining (used in our study) are considered AI-hard. Consequently the generated reports lack coherence, which is vital when results are intended to be used in real environments. We believe that this should become manageable with further improvement in NLP performance but also with the introduction of human workflows (e.g. moderation) or human-in-the-loop AI paradigms.

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